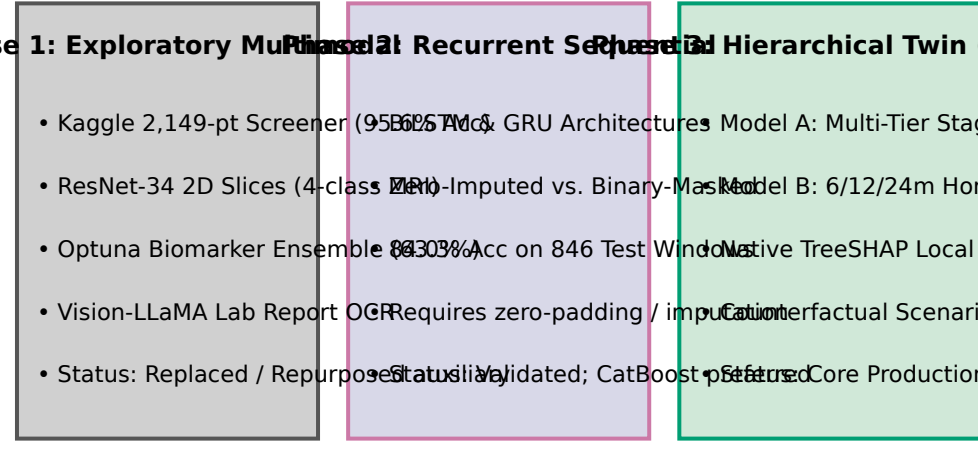
Phase 1: Exploratory Multimodal
• Kaggle 2,149-pt Screener (95.6% AUC)
• ResNet-34 2D Slices (4-class Dx)
• Optuna Biomarker Ensemble (83.3% Acc)
• Vision-LLaMA Lab Report OCR
• Status: Replaced / Repurposed
Phase 2: Recurrent Sequential
• LSTM & GRU Architectures
• Zero-Imputed vs. Binary-Masked
• 84.03% Acc on 846 Test Windows
• Requires zero-padding / imputation
• Status: Validated; CatBoost retained
Phase 3: Hierarchical Twin (Final)
• Model A: Multi-Tier Stager (T1-T4)
• Model B: 6/12/24m Horizon Regressors
• Native TreeSHAP Local Attributions
• Counterfactual Scenario Engine
• Status: Core Production Engine